GLYTHERMIN® P35

Powerful Cooling. Zero Labels. Total Peace of Mind: GLYTHERMIN® P35

As data centres evolve to support high-performance computing, AI workloads, and cloud infrastructure, the demand for efficient, sustainable, and scalable cooling solutions has intensified.

With decades of experience in thermal management and a globally recognized portfolio of high-performance coolants, BASF brings deep technical know-how and formulation expertise to the evolving demands of data centre cooling. Leveraging its proven **GLYTHERMIN®** heat transfer technologies, BASF supports direct-to-chip cooling applications with a solution that offer long-term corrosion protection and thermal stability. Tailoring its offerings to the specific needs of hyperscale and edge data centres, **GLYTHERMIN® P35** ensures operational reliability in even the most demanding environments.

GLYTHERMIN® P35 is a Propylene Glycol based heat transfer fluid with an Organic Acid Technology (OAT) inhibitor package. The fluid offers a compelling alternative to traditional glycol-based coolants, particularly in direct-to-chip (D2C) and hybrid liquid cooling systems.

GLYTHERMIN® P35 is available as concentrate, which reduces shipping costs and minimizes environmental impact. This makes our product more cost-effective and supports sustainable practices.

Properties	Unit	GLYTHERMIN® P35
MPG Concentration		Concentrate
Appearance		Clear to slightly hazy
Density at 20°C	g/cm^3	1.052 – 1.057
Refractive Index at 20°C		1.434 – 1.438
pH - value		8.0 – 9.0
Water content	%	4
Flash point	°C	>100
Reserve alkalinity	ml	8 – 12

Superior Thermal Performance

GLYTHERMIN® P35 delivers excellent thermal conductivity and heat capacity, enabling efficient heat removal from high-density server racks. It maintains stable performance across a wide temperature range and is particularly effective in D2C applications where precise thermal control is critical.

Long-Term Corrosion Protection

OAT inhibitors form a protective layer on metal surfaces without depleting over time, offering long-lasting corrosion resistance for materials like copper, aluminium, stainless steel, and brass. This extends the lifespan of cooling systems and reduces maintenance frequency.

Environmental and Regulatory Compliance

As an OAT formulation, **GLYTHERMIN® P35** is free from nitrites, borates, phosphates, silicates and 2-EHA – substances often flagged for environmental or health concerns. **GLYTHERMIN® P35** is classified as non-hazardous and the main components are biodegradable, aligning with sustainability goals and regulatory requirements.

Reduced Maintenance and Downtime

Unlike traditional fluids that require regular replenishment of inhibitors, OAT-based fluids are depletion-free. This reduces the need for frequent fluid analysis and top ups, lowering operational costs and minimising system downtime.

Material Compatibility and suitability

GLYTHERMIN® P35 is compatible with a wide range of elastomers and thermoplastics used in data centre cooling systems. This makes it suitable for both new installations and retrofits in existing infrastructure.

The new product fulfills the requirements of ASTM D3306 (international standard for engine coolants) for propylene glycol-based engine coolant concentrates and ASTM D8039 (international standard for heat transfer fluids)

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